Anhalt University of Applied Sciences
Biomedical Engineering (M.Sc.)
(Department of Electrical, Mechanical and Industrial Engineering)

Imaging

Lab Report : 3

Student Name: Agbadi samson komlan **Student ID:**

Lecturer:

Prof. Dr. Marianne Maktabi

Q1. Difference Between RGB Image and Grayscale Image (Array Perspective)

1. RGB Image Representation

The RGB image used (`ME.jpg`) is represented as a 3 dimensional array with the shape:

H×W×3

where:

- **H** = height of the image
- **W** = width of the image
- **3** = the three color channels (**Red, Green, Blue**)

Each pixel in an RGB image contains **three values**, corresponding to the intensity of the Red, Green, and Blue channels.

Key Points

- Each pixel is represented as a **vector of three values**.
- The image contains **three channels**.
- The array shape depends on:
  - height,
  - width,
  - and the number of color channels.

2. Grayscale Image Representation

After conversion to grayscale, the image becomes a 2 dimensional array with the shape:

H×W

Each pixel now contains only one value, which represents the intensity (brightness) of the image.

Key Points

- Only one value per pixel is stored.
- There are no color channels anymore.
- The grayscale image represents brightness only.
- The image dimension is reduced from 3D to 2D.

Q2. Modifiable Parameters of Histogram

The main modifiable parameters are:

1. **Number of bins**
2. **Value range**
3. **Image representation format**

1. Number of Bins

The number of bins determines how intensity values are grouped in the histogram.

Increasing the Number of Bins

- Increases the resolution of the intensity distribution.
- Shows more details in the histogram.
- However, it may introduce:
    - noise,
    - sparsity,
    - and irregular peaks.

Decreasing the Number of Bins

- Produces a smoother histogram.
- Reduces noise.
- However, some fine details may be lost because multiple intensity values are grouped together into broader categories.

2. Value Range

The range parameter must match the image data type.

Examples:

- **0 to 1** → for normalized images
- **0 to 255** → for 8-bit images

If the range does not match the image type, the histogram may become distorted or misleading.

Q3. Comparison Between Histogram Plots

PIL Histogram

- Pixel intensity range: **0 to 1**

OpenCV Histogram

- Pixel intensity range: **0 to 255**

Matplotlib Histogram

- Pixel intensity range: **0 to 1**

Difference Between RGB Histogram and Grayscale Histogram

The histogram of an RGB image contains **three separate channel distributions** corresponding to the:

- Red channel,
- Green channel,
- Blue channel.

This is because an RGB image is represented as a **3D array**:

$H \times W \times 3$

On the other hand, the grayscale image histogram contains only a **single intensity distribution** because the grayscale image is represented as a **2D array**:

$H \times W$

with only one channel.

Q4. Preferred Bin Size

A moderate bin size such as **64** or **256** is preferable because it balances detail and smoothness.

For this work, **256 bins** were selected for the 8-bit image because:

- pixel intensities naturally range from **0 to 255**,
- giving exactly **one bin per intensity value**.

This means:

- all intensity details are preserved,
- no information is hidden,
- and the histogram accurately represents the image distribution.

Therefore, 256 bins provide the most detailed and accurate representation for an 8-bit image.

What the Histogram Reveals About the Image

From the histogram analysis, the following observations were made:

1. There is a major spike near intensity value **1** (white region).
2. The **blue channel dominates heavily** toward the right end of the histogram.
3. The image has **limited mid-tone variability**, meaning:
   - most pixels are concentrated at either:
     - very dark values,
     - or very bright values.

This indicates that the image contains strong contrast with fewer medium-intensity regions.